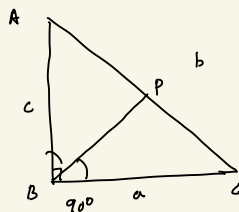


Q3

UGB
Mock
Toppers.

$$\left. \begin{array}{l} AP^2 = ar(\triangle ABC) \\ \text{To show: } AB = BC \end{array} \right\}$$

Some possible variants:

i) Given that $AP \cdot PC = ar(\triangle ABC)$

To show: $AB = BC$

ii) Given that: $BP^2 = ar(\triangle ABC)$

To show: $AB = BC$.

Q. 1.a, b $\in \mathbb{N}$ are distinct, ab is a perfect sq.If $a+b$ is prime, it must be of the form $4k+1$.Result:Let p be a prime of the form $4k+3$,

and $p \mid a^2 + b^2$ ($a, b \in \mathbb{Z}$).

Then $p \mid a$, $p \mid b$.

App:1) Show that, there cannot exist any integer triple (x, y, z) s.t.

$$\left. \begin{array}{l} x^2 + y^2 = 11z^2 \\ \text{and } x+y+z = 10000 \end{array} \right\}$$

hold simultaneously.

Pf:

$11z^2 = x^2 + y^2 \Rightarrow 11 \mid x^2 + y^2$

$\Rightarrow 11 \mid x, 11 \mid y$ ($\because 11$ is

$\Rightarrow x = 11k_1, y = 11k_2$ a prime of the form $4k+3$)

$(k_1, k_2 \in \mathbb{Z})$

Then, $x^2 + y^2 = 11z^2$

$\Rightarrow (11k_1)^2 + (11k_2)^2 = 11z^2$

$\Rightarrow 11(k_1^2 + k_2^2) = z^2 \Rightarrow 11 \mid z^2$

$\Rightarrow 11 \mid z$ ($\because 11$ is prime)

$\Rightarrow 11 \mid x+y+z$

but $11 \nmid 10000$!!

2) Show that there cannot exist $a, b, c, d \in \mathbb{N}$ s.t

$$a^2 + b^2 = 2027(c^2 + d^2) \leadsto \text{prime}$$

$\hookrightarrow p \equiv 3 \pmod{4}$

$$p \mid a^2 + b^2 \Rightarrow p \mid a, \quad p \mid b \quad (p \text{ is prime } \equiv 3 \pmod{4})$$

$$\Rightarrow \exists a = a_1 p, \quad b = a_2 p.$$

$$\text{Now, } (a_1 p)^2 + (a_2 p)^2 = p(c^2 + d^2)$$

$$\Rightarrow p(a_1^2 + a_2^2) = p(c^2 + d^2)$$

$$\Rightarrow \frac{p \mid a_1^2 + a_2^2}{p \mid c^2 + d^2} \Rightarrow p \mid c, \quad p \mid d$$

$$\Rightarrow c = p c_1, \quad d = p c_2$$

$$p(a_1^2 + a_2^2) = p^2(c_1^2 + c_2^2) \Rightarrow a_1^2 + a_2^2 = p(c_1^2 + c_2^2)$$

$$p \mid a_1^2 + a_2^2 \Rightarrow p \mid a_1, \quad p \mid a_2$$

$$\Rightarrow a_1 = p a_2, \quad a_2 = p a_3$$

$$a > a_1 > a_2 > a_3 > \dots$$

$\Rightarrow$
natural no.

By infinite descent method
no such a can exist.

1st Result: Let p be an ^{odd} prime s.t. $\exists n \in \mathbb{Z}$ s.t.

$$\frac{p \mid n^2 + 1}{\sim} \left(\begin{array}{l} x^2 \equiv -1 \pmod{p} \\ \text{has a soln in} \\ \text{integers} \end{array} \right)$$

Then p must be of the
form $4k+1$.

Pf:

We have $p \mid n^2 + 1 \Rightarrow p \nmid n$ $\left(\begin{array}{l} p \nmid n \\ \Rightarrow p \nmid n^2 \\ \Rightarrow p \nmid 1 \end{array} \right)$
 $\Rightarrow n^2 \equiv -1 \pmod{p}$
 So, p, n are coprime.

Assume, on the contrary, that

$$p \equiv 3 \pmod{4}$$

$$\Rightarrow p = 4k+3 \quad (k \in \mathbb{N} \cup \{0\})$$

By FLT

$$n^{p-1} \equiv 1 \pmod{p}$$

$$\Rightarrow n^{4k+2} \equiv 1 \pmod{p}$$

$$\Rightarrow (n^2)^{2k+1} \equiv 1 \pmod{p}$$

$$\Rightarrow (-1)^{2k+1} \equiv 1 \pmod{p}$$

$$\Rightarrow -1 \equiv 1 \pmod{p}$$

$$\Rightarrow p \mid 2 \quad !!$$

Hence, our assumption about $p \equiv 3 \pmod{4}$
is wrong. Thus p has to be of the form $4k+1$.

Main Result:

Let p be a prime of the form $4k+3$.

If $p \mid a^2 + b^2$, ($a, b \in \mathbb{Z}$), then

$$p \mid a, \quad p \mid b.$$

Pf:

Assume, on the contrary, that $p \nmid a$.
Then, $\exists x_0 \in \mathbb{Z}$ s.t.

$$ax_0 \equiv 1 \pmod{p}$$

$$\Rightarrow a^2 x_0^2 \equiv 1 \pmod{p}$$

$$\Rightarrow -b^2 x_0^2 \equiv 1 \pmod{p}$$

$$\Rightarrow p \mid (bx_0)^2 + 1$$

$$\Rightarrow p \equiv 1 \pmod{4}$$

Contradiction!

So, $p \mid a$ and in the similar manner,

$$p \mid b. \quad \square$$

$$(13 = 9 + 4)$$

$$, \quad 29 = 25 + 4)$$

Fermat's Christmas Thm: Any prime of the form $4k+1$

can be expressed as a sum of two perfect squares.

Converse of the 1st Result: Let p be a prime of the form $4k+1$. Then $\exists n \in \mathbb{Z}$ s.t.

$$p \mid n^2 + 1.$$

$$\begin{aligned} & \text{(Wilson: } \textcircled{4k} \text{)} \\ & \quad \downarrow \\ & \quad 12! \equiv -1 \pmod{13} \\ & \quad \downarrow \\ & \textcircled{2k} \rightarrow (1 \times 2 \times \dots \times 6) \times (7 \times \dots \times 12) \equiv -1 \pmod{13} \\ & \quad (1 \times 2 \times \dots \times 6) \times (-6) \times (-5) \times \dots \times (-1) \equiv -1 \pmod{13} \\ & \quad \downarrow (-1)^{2k} \\ & \Rightarrow (6!)^2 \equiv -1 \pmod{13} \\ & \Rightarrow 13 \mid (6!)^2 + 1 \\ & \quad \textcircled{2k} \rightarrow \textcircled{4k} \end{aligned}$$

$$\begin{aligned} [5 \mid 2^2 + 1] \\ [13 \mid 5^2 + 1] \\ [17 \mid 4^2 + 1] \end{aligned}$$

$$p \mid \left[\left(\frac{p-1}{2} \right)! \right]^2 + 1$$

$\uparrow$
 $\textcircled{n'}$

$$\text{Let } p = 4k+1 \quad (k \in \mathbb{N})$$

By Wilson

$$(p-1)! \equiv -1 \pmod{p}$$

$$\Rightarrow (4k)! \equiv -1 \pmod{p}$$

$$\Rightarrow [1 \times 2 \times \dots \times 2k] [(2k+1) \dots (4k)] \equiv -1 \pmod{p}$$

$$\Rightarrow 1 \times 2 \times \dots \times 2k [(-2k)(-2k+1) \dots (-1)] \equiv -1 \pmod{p}$$

$$\Rightarrow (1 \times 2 \times \dots \times 2k) \cdot (2k)! \equiv -1 \pmod{p} \quad (p = 4k+1)$$

$$\Rightarrow ((2k)!)^2 \equiv -1 \pmod{p}$$

$$\Rightarrow p \mid ((2k)!)^2 + 1$$

Here, such n exists. $\square$.

$\square$ Let p be an odd prime, then the quadratic congruence

$$x^2 \equiv -1 \pmod{p}$$

will have an integer solution

if and only if $p \equiv 1 \pmod{4}$.

$$\square \text{ Let, } ab = k^2, \text{ let, } \underline{a+b} = p \Rightarrow \boxed{0 < a < p, \quad 0 < b < p}$$

$$\underline{p^2 = (a+b)^2 = (a-b)^2 + 4ab = (a-b)^2 + 4k^2} = \underline{(a-b)^2 + (2k)^2}$$

$$\text{AOTC, } p \equiv 3 \pmod{4}$$

$$\boxed{-p < \underline{a-b} < p} \quad \begin{matrix} \rightarrow \\ \text{and } p \mid (a-b) \end{matrix} \quad \begin{matrix} b \mid a-b \\ b \mid 2k \end{matrix}$$

$$\Rightarrow a-b=0$$

$$\Rightarrow a=b \quad !!$$

(a, b are distinct)